SMART INVOICE CHASING

Revised Implementation Roadmap & Bootstrap Plan

A prioritized plan incorporating the uploaded Bootstrap Recommendation and review notes.

Included: navigation, financial truth, invoice import and payment verification, TDS, safe email collections, AI Rules, sender configuration, privacy, security, testing, staging and customer validation.

Excluded: the approved intro slides and sign-in/sign-up flows. Do not revisit them unless a concrete regression is evidenced and separately approved.

Status: planning document only. It does not claim any task has already been implemented or any integration has been verified.

Prepared 24 October 2026

1. Executive Decision

Build a small, honest core; charge a small amount early; spend only when a specific trigger is reached. The revised order places financial correctness before live sender setup and AI autonomy.

Default decision: start without n8n. Scheduling, retries and email dispatch can initially be handled through application-controlled jobs/queues (for example, Supabase scheduled jobs or Vercel Cron, after checking plan limits). Reconsider n8n only if a concrete later need justifies its operational, security, licensing and hosting cost.

Decision	Initial direction
Automation orchestration	No n8n in the core MVP; revisit later for low-risk owner alerts if useful.
Collection channel	Email first, after sender authentication and safety controls are tested.
WhatsApp	Defer API integration. Consider click-to-chat and manual reply logging only if pilot customers need it.
SMS	Defer; show “Not available yet” until legal/provider prerequisites are verified.
Payments	Manual verification with reference/UTR and evidence; bank-statement CSV matching later.
Billing integration	Manual UPI/bank transfer and invoice initially; consider billing automation after roughly 10 paying customers.
AI autonomy	Keep limited; deterministic rules and human approval govern collection actions.
Release	No push/deploy/provider activation or real collection campaign without explicit approval.

2. Correct Scope and Guardrails

Included product areas

- Sidebar and route integrity; dashboard; invoices/import; collections; receivables aging; customers; reports; risk matrix; AI decisions/human review; activity; workspace management; profile; AI Rules and all Settings tabs.
- Real account identity, workspace business identity, email/mobile sender configuration, provider status, safe dispatch, payment verification and privacy/security requirements.
- Staging, automated tests, backups, monitoring and pilot measurement.

Explicitly excluded

- The approved seven intro slides.
- Sign-in, sign-up and authentication redesign. Treat as completed unless a concrete regression is found and separately approved.
- SMS sending, WhatsApp Business API, n8n, AI autonomy and billing integration in the initial core MVP.
- Unrelated redesigns, new packages without justification, and broad refactors during narrow tasks.

Repository safeguards

- Repository: D:\solidfut\Project code files\saas
- Production: https://smart-invoice-chasing.vercel.app
- GitHub: https://github.com/Harshith-SR1/smart-invoice-chasing
- Before each task, inspect current Git status and preserve all pre-existing tracked and untracked work. Never reset, stash, revert, overwrite or delete existing work to clean the tree.
- Use a dedicated work branch and make frequent local commits when appropriate to protect progress; push and deploy remain gated by explicit approval.

3. Revised Build Order

Phase 0 — Cheap trust fixes and route integrity

- Finish and verify the agreed sidebar restoration; preserve the top navbar and profile dropdown.
- Fix Configure AI Rules so it opens the exact AI Rules configuration, not only the generic Settings landing page.
- Audit every sidebar link, direct route, refresh, active state, mobile drawer and keyboard focus.
- Replace misleading integration/plan/automation labels with honest “Not connected”, “Not configured” or “Not available yet” states until verified.
- Remove unintended demo data only after checking whether values belong to the actual authenticated workspace.

Phase 1 — Financial truth and core invoice lifecycle

- Implement robust CSV/XLS/XLSX import: preview, header mapping, normalization, row validation, rejected-row reasons, duplicate strategy and workspace-scoped persistence.
- Make new workspaces genuinely empty; separate loading, empty, error and populated states; remove mock-data flashes.
- Define and implement server-side invoice totals, outstanding balances, overdue amounts, recovered totals and recovery-rate calculations.
- Only verified payments reduce outstanding balances or count as recovered. A customer statement “I paid” is a claim, not proof.
- Manual payment verification MVP: mark received with amount, date, UTR/reference and evidence; record who verified it and when.
- Partial payments reduce balance only by the verified amount. Fully settled invoices stop collection reminders.
- Add TDS/short-payment adjustment support as a distinct, auditable adjustment type, with amount, reason, applicable invoice and certificate tracking where relevant. Do not silently treat it as cash received; validate the accounting/tax workflow with a CA.
- Plan bank-statement CSV matching as a later phase; no gateway integration initially.

Phase 2 — Rules-based Chase Today and collection safety

- Implement a deterministic eligible-action list based on invoice due dates, verified balances, payment state, dispute/hold state, promises, workspace pause and previous send history.
- Keep WAIT, NUDGE, FOLLOW_UP, ESCALATE, HUMAN and STOP consistent across Dashboard, invoice detail, Collections and AI Decisions.
- Add dry-run preview early: show intended recipient, message, rule, next action and why an invoice will be sent or skipped, without dispatching.
- Add a global sending kill switch, per-workspace daily send caps, quiet hours and duplicate prevention.
- Re-check eligibility immediately before dispatch. Stop if paid, held, disputed, paused, stopped or waiting for required approval.
- Add a human-review gate for legal/direct-phone escalation and other sensitive actions.
- Track promise recorded date separately from promised payment date; a promise is not a payment.

Phase 3 — Reply logging and email-first delivery

- Make email the first live channel only after financial and safety tests pass.
- Configure the real sender account/domain; do not assume a Gmail address or phone number is a working sender.
- Verify sender domain authentication: SPF, DKIM and DMARC alignment as applicable to the chosen provider.
- Implement a reply-to strategy and an inbound reply address if supported; include a manual “Log reply” action as the MVP fallback.
- Track message states truthfully: drafted, queued, attempted, sent, delivered, bounced, complained, failed or suppressed, based on actual evidence.
- Implement bounce/complaint handling, a suppression list and an opt-out mechanism before real campaigns.
- Decide shared sending domain versus per-workspace domains; assess the reputation blast radius of shared sending.
- Add provider connection test, test-send, last verification, reconnect and visible failure states.
- For Gmail OAuth, verify current provider app-verification and consent requirements before committing to the implementation.

Phase 4 — AI Rules and constrained assistance

- Only after core financial truth and safe sending work, connect configured cadence and tone to the deterministic decision engine.
- Validate cadence thresholds and prevent conflicting/duplicate schedules.
- Enforce receiver pause/resume, promise pause, disputes/holds, STOP, HUMAN review and approval rules on the server.
- AI may classify replies or draft messages; it cannot verify payment, change financial truth or bypass send policy.
- Show “Why did this action happen?” with the rule, evidence, next action and approval status.
- Only show saved/synced after confirmed database success; surface failures and permission errors.

Phase 5 — Other pages and workspace settings

- Collections: accurate Promisers, Disputers, Ghosters and Active classifications; pause/resume audit trail and consistent next actions.
- Aging: Not due, 1–7, 8–30, 31–60, 61–90 and 90+ day buckets; explicitly document date boundaries and reconcile totals.
- Customers: real records and verified history; show insufficient history rather than invented risk.
- Risk Matrix: evidence-based indicators with explanations and limitations.

- Reports: distinguish promise, approved, attempted, sent, delivered, failed and recovered outcomes.
- Activity: full timestamps, timezone consistency and event history backed by real provider/application events.
- Workspace Management: correct identity, members, roles, invitations, seats and plan; no unverified demo data.
- Settings tabs: Business, AI Receiver, Collection Rules, AI Autonomy, Notifications, Email Service, Security and Preferences; each must persist and affect actual behaviour.
- User profile, workspace identity, sender identity and recipient identity remain separate data categories.

Phase 6 — Privacy, security, staging and operations

- Use a separate Supabase project for staging and Vercel preview deployments; never use production customer data for routine test runs without an approved safe process.
- Verify Row Level Security (RLS), server-side authorization, workspace isolation, role checks, rate limits, secret storage and signed webhooks.
- Add retention/deletion, data export, breach response and customer data handling processes; plan for a data processing agreement and professional review of applicable Indian privacy obligations.
- Choose and verify hosting region based on product needs, contractual promises and applicable legal advice; do not claim a region or compliance posture without evidence.
- Set up backup/restore testing, error monitoring, uptime checks and provider failure alerts.
- Add Continuous Integration (CI) for type checks, tests and production build on every push to the work branch.
- Add automated cross-workspace access tests and known-total financial tests.

Phase 7 — Pilot, learn and reinvest

- Pilot with 5–10 businesses for six weeks; charge a small amount early where feasible (the research report suggests testing willingness to pay, even at ■499).
- Measure days to payment, reply rate, time saved, verified recovered amount and willingness to pay.
- Track weekly: businesses that imported invoices, businesses that sent a reminder, and verified recovered amount per customer.
- Set a failure criterion before the pilot. Example from the research: if fewer than 3 of 10 pilot users are willing to pay after six weeks, reassess direction.
- Speak with 3–4 Chartered Accountants (CAs) and have professional review of tax, privacy, terms, data processing and business setup.

4. Financial Acceptance Rules and Test Fixtures

Scenario	Expected outcome
■1,00,000 invoice; ■40,000 verified payment	Outstanding = ■60,000.
■1,00,000 invoice; customer says “paid” but no verification	Outstanding remains ■1,00,000; claim is logged separately.
■1,00,000 invoice; ■25,000 verified partial payment	Outstanding = ■75,000.
Fully verified settlement	Outstanding = ■0; no further collection reminders.
Disputed or held invoice	No dispatch while the applicable hold/dispute condition remains active.
Promise recorded	Do not count as recovered; track promise date separately.
TDS adjustment	Record as a distinct auditable adjustment; do not falsely record as cash received.
Repeated job/webhook event	No duplicate payment, reminder or financial event.

Also test Indian currency/number formatting (including lakhs/crores), date boundaries for aging, timezone handling and cross-workspace access isolation.

5. Email and Messaging Boundaries

Channel	MVP decision	Prerequisites / limits
Email	First live channel after core gates	Authenticated sender/domain, SPF/DKIM/DMARC as applicable, test-send, bounce/complaint handling, suppression and opt-out.
WhatsApp	Defer Business API	If demanded by pilots, consider click-to-chat plus manual reply logging first; API needs provider setup, business verification/templates and confirmed pricing.
SMS	Not available yet	Defer until relevant Indian DLT registration, sender/template approvals and provider requirements are verified.

The source report flags Indian telecom DLT and WhatsApp business/template prerequisites. Treat this as a research finding to verify with primary authorities and a provider before relying on it. This plan does not provide legal advice.

6. n8n Decision Record

Decision

Do not make n8n a dependency of the initial MVP. The core needs scheduling, retries and email API calls, which may be handled by application-controlled scheduled jobs and a send-queue table. Verify current plan limits and operational needs before selecting the implementation.

Why defer

- The locally recorded n8n URL (127.0.0.1:31415) is not reachable by the production service as a local address.
- A production-hosted n8n adds hosting, patching, security and operational work.
- The workflow repository is untrusted input: inspect every node, external call, credential use and licence before reuse.
- Do not put customer OAuth credentials into n8n without confirming licensing and security implications; prefer n8n calling the application's authenticated API.
- Verify current n8n version, security advisories and patch level if reconsidered; verify webhook signatures in the application.

Reconsider only if

- The core product is stable and paying customers create a real orchestration need.
- A workflow demonstrably saves engineering/operational time versus application-managed jobs.
- Hosting, licensing, credential boundaries, patching, monitoring and failure recovery have been reviewed.
- Initial candidate use case is low risk, such as owner notifications—not financial calculations or payment verification.

7. Bootstrap Spending Triggers

Trigger	Suggested spend/action	Reason
Real customer financial data enters	Move to a suitable paid database plan after confirming current price and terms.	Avoid unacceptable pausing/backup risk for customer records.
First paying customer	Move off any hosting tier whose terms prohibit commercial use; verify current terms.	Remain within provider terms.
Email volume exceeds free limits or second domain is needed	Upgrade email plan or choose suitable provider.	Avoid blocked sends and support required domains.
Before first paying customer	One-time CA/lawyer review.	Terms, privacy policy, data processing agreement, GST and business structure.
Customers repeatedly request WhatsApp	Evaluate provider and unit economics; pass usage costs through where appropriate.	Protect margin.
Revenue is steady and operational need is proven	Add monitoring and reconsider n8n/SMS only if justified.	Avoid premature spend.

The uploaded report estimates a paid stack in the range of US$50–70/month and around 5–6 customers at ■999/month to cover it. This is an estimate, not a current quote or guarantee. Recheck pricing, usage limits, backups and commercial terms before committing.

Reinvestment rule: revenue first covers infrastructure, then the next justified expense. Founder payments should follow a pre-agreed financial threshold.

8. Privacy, Legal and Business Readiness

- Draft Terms of Service, Privacy Policy, data processing agreement, opt-out process, data retention/deletion process, data export procedure and breach-response plan.
- Document who is responsible for customer/end-customer data and which processors receive it.
- Have a qualified lawyer review applicable Digital Personal Data Protection (DPDP) obligations and timelines; the uploaded report's legal timeline is a research conclusion, not legal advice.
- Ask a CA about business structure, founder roles/equity/code ownership, GST applicability, TDS treatment and the applicability of MSME/45-day messaging claims.
- Verify payment gateway onboarding requirements before adding a gateway; initial customer payment can be by UPI/bank transfer with manually issued invoices if professionally appropriate.
- Set up support email, onboarding guide, ownership/role responsibilities and a support channel.
- Verify domain ownership and sender DNS records for smartchasing.io before sending.

9. Staging, CI and Verification Checklist

1. Separate staging Supabase project and Vercel preview environment exist and use test data.
2. CI runs type checks, unit/integration tests and production build on every push to the work branch.
3. Known-total tests pass, including ■1,00,000 – ■40,000 verified = ■60,000.
4. Cross-workspace tests prove one workspace cannot read or mutate another workspace's records.
5. Import tests cover valid/invalid rows, duplicate handling and persistence.
6. Payment tests cover claim vs verified, full/partial settlement, TDS adjustment and duplicate events.
7. Collection tests cover due/overdue cadence, quiet hours, daily caps, global kill switch, dry-run and duplicate prevention.
8. Email tests cover test-send, invalid recipient, bounce, complaint, suppression, opt-out, provider outage and reconnect.
9. Settings tests prove persistence and actual behavioural effect after refresh.
10. n8n is not required for the MVP; if added later, version/licence/security review and webhook verification are documented.
11. Backup restore and monitoring/alert tests are complete before production customer reliance.

10. Antigravity Execution Protocol

1. **Inspect first:** report current Git status, target files, relevant routes/functions and tests. Preserve all existing tracked changes and untracked artefacts.
2. **State the task:** identify one defect, acceptance criteria and expected files before editing.
3. **Implement only that scope:** avoid unrelated refactoring and unnecessary dependencies.
4. **Verify:** run targeted tests, `npx tsc --noEmit`, `npm run build` and relevant security/financial tests where feasible.
5. **Report:** list files changed, tests/results, evidence, blockers and next task. Do not claim a provider or production path works without testing it.
6. **Protect work:** use local commits on a dedicated work branch where appropriate; no push, deployment or provider activation without explicit approval.

11. Recommended Next Actions

1. Finish and verify sidebar restoration.
2. Fix Configure AI Rules navigation and honest connection/status labels.
3. Implement and test financial truth, import persistence, manual verified payments and TDS adjustment handling.
4. Build rules-based Chase Today, dry-run, kill switch, caps, quiet hours and duplicate prevention.
5. Add manual reply logging and safe email-first sending with domain authentication, opt-out and bounce/complaint suppression.
6. Wire AI Rules into the verified state and constrained action pipeline.
7. Complete workspace/settings consistency, privacy/security, staging and CI.
8. Run a six-week pilot with 5–10 businesses; evaluate payment willingness and product outcomes before adding n8n, SMS, WhatsApp API or billing integrations.

12. Definition of Done

The product is ready for a pilot only when an invoice can be imported, its balance remains correct, a payment is manually verified with evidence, the rules-based action is explainable, an eligible reminder can be previewed safely, sending can be stopped globally, replies can be logged, and all outcomes are auditable. Live email requires a real authorised sender and deliverability/suppression safeguards. Production readiness additionally requires staging, security, backup/restore and end-to-end evidence.

Excluded from this plan: the approved intro slides and sign-in/sign-up flows. Do not revisit them without a concrete regression and separate approval.

Research caution: the Bootstrap Recommendation contains vendor pricing, licence and regulatory findings based on sources available on 24 October 2026. Confirm current provider terms/pricing and primary-source legal/regulatory requirements before relying on them.